

Solution, simulation and analysis of the angular dynamics of a free body with a magnetic moment subject to a rotating magnetic field

Abstract/Research Context

This application is a symbolic, numerical, and visual analysis framework for studying the angular dynamics of free or magnetically bounded bodies subjected to rotating magnetic fields. Magnetic bounding by rotating fields is a relatively new phenomenon in which a rigid body with a magnetic moment can be trapped at a precise location by a rotating magnetic field with a gradient, without requiring a local minimum of magnetic potential. The effect is essentially realizable with dipole magnets. In this interaction, the body couples to the field through its magnetic moment, which is fixed in the body frame, and may undergo translational and angular oscillations of typically small amplitudes. These oscillators allow the body to find a quasi-stable equilibrium with driving field and the confinement mechanism which imposes within magnetostatic except using diamagnetism.

This work focuses on the angular part of the dynamics, which is responsible for keeping the body in proper alignment with the rotating field. In a basic model, the angular equation consists of a symmetric conical motion of the body's magnetic moment around the axis of the rotating field synchronized with this field. This alignment mechanism can be associated with the phase-lag property of driven oscillators: in this case the phase of the body's magnetic moment is antiparallel to the field in the azimuthal components, while parallel alignment occurs in the zenithal components of the magnetic moment and the rotating field.

Since antiparallel alignment results in repulsion and parallel in attraction, the stability in translational degrees of freedom is obtained by the system ability to adjust these opposing effects by variation of the zenithal angle of the body -- which, in turn, corresponds to the amplitude of this driven oscillation as a function of the strength of magnetic torque acting on the body.

As the body moves closer in the field gradient, the stronger field increases the magnetic torque, causing the oscillation amplitude to increase. This corresponds to a larger zenithal angle and makes the antiparallel alignment more pronounced. Consequently, the repulsive component grows faster than the attractive component, producing an effectively positive translational stiffness.

A related self-regulating mechanism can also arise from translational oscillations, especially in the radial plane with respect to the rotation axis of the driving field in a larger central frame and makes the antiparallel alignment more pronounced. In this rotating field frame, the explicit time dependence cancels and the equations of motion become autonomous. These equations, supported by numerical simulations and experimental observations, provide the operating points for local linearization. The equations of motion are then expanded to first order about the equilibrium state defined by the generalized coordinates and their first derivatives, yielding the state-matrix representation used for eigenvalue analysis.

The eigenvalue analysis exposes the local stability characteristics and natural angular frequencies of the synchronized solution. In the absence of damping, the system can be marginally stable, with eigenvalues whose real parts are zero. In the presence of damping, stable solutions become asymptotically stable when the clear parts of all eigenvalues are negative. When the field inclination angle γ is reduced below the required stability threshold, one real part becomes positive, clearly indicating an unstable regime.

The simulations and eigenvalue results support each other in identifying stability limits and reproducing accurately the observed eigenfrequencies. The imaginary parts of the eigenvalues give the natural angular frequencies of the synchronized solution. In the stable regime, two principal eigenfrequencies are typically observed: one associated with the driving frequency, and the other slightly above it. The separation of these frequency branches is mainly controlled by the field inclination angle γ , while increasing oscillation amplitude tends to shift both frequencies downward. This downward shift can be interpreted as a softening-type nonlinear effect. Through parameter sweeps, the utility makes it possible to observe frequency branching, stability boundaries, damping effects, and the influence of inertia ratios.

Simulations further indicate the existence of excited bounded modes within the stable regime. In these modes, the zenithal angle of the magnetic moment can vary over a large range, and the corresponding state-space trajectories may form cycloidal, rosette-like, looped, or otherwise structured patterns. In bounded cases, these trajectories can converge toward attractors or preferred bounded regions in state space, indicating stable excited forms of motion in addition to the basic synchronized equilibrium. The built-in state-variable plots, phase portraits, and polar plots provide a practical way to examine these modes and compare them with regular synchronized motion, beat formation, nonlinear amplitude limitation, and the onset of instability.

Section C analyzes the behavior of the state variables in the presence of damping. In particular, it is shown that, in the absence of electromagnetically induced spin-up torques, θ converges to $-\pi \cos \theta$. This condition corresponds to a zero z-component of the body's angular velocity in local coordinates and provides a useful reference state for setting up damped simulations.

Direct nonlinear simulations and eigenvalue results support each other in identifying stability limits and reproducing the observed eigenfrequencies. Simulations also reveal an angular instability mechanism when the body has a small non-axisymmetry, which is often encountered in experiments. In such cases, the nonlinear dynamics of the system can use this asymmetry to promote resonant angular modes.

Section E reformulates Ucar (2021)'s semi-empirical stability formula in terms of the driving-field elevation angle γ . The formula is applicable to isotropic bodies under the condition $\psi = -u$ and is found to predict stability limits in agreement with both eigenvalue analyses and direct simulations.

A central purpose of the application is to support the investigation of magnetic bound-state dynamics beyond simple numerical simulation. The symbolic derivation of the equations of motion allows the assumptions, coordinate definitions, and model reductions to be inspected directly. The numerical solver enables time-domain simulations, while the animation, plotting, and phase-portrait tools help visualize the physical motion and the evolution of state variables.

The application also includes linearization and eigenvalue analysis tools in Section D. These are useful for studying local stability around synchronized operating states, identifying natural angular frequencies, and comparing predicted small-signal behavior with frequencies observed in full nonlinear simulations. By sweeping parameters such as field rotation rate, field inclination, damping, magnetic torque strength, inertia ratios, and initial conditions, the user can observe how stable, unstable, quasi-synchronized, and transitional regimes emerge.

Therefore, this utility can be used as both a research notebook and an exploratory computational laboratory for angular magnetic dynamics. It may be useful for studying magnetically bounded motion, rotating-field stabilization, resonance-like behavior, damping-dependent equilibrium states, non-axisymmetric instability mechanisms, and the relationship between nonlinear simulations and linearized eigenvalue spectra.

► A. Quick Guide

► A. Derivation of Equations of Motion

Equations of angular motion of a body subject a dynamic magnetic torque are calculated in body coordinates as follow. Equations generated by this flow are used in numeric integration solver in order generate simulation data.

1. Symbols and Generalized Coordinates

$$\omega, I, mB, \gamma, \zeta, t \in \mathbb{R}, \quad \theta(t), \phi(t), \psi(t), \quad \frac{\partial \theta}{\partial t}, \dot{\phi} = \frac{d\phi}{dt}, \dot{\psi} = \frac{d\psi}{dt}.$$

Here ω is the angular velocity of the rotating field. I the diagonal moment-of-inertia tensor with components (I_1, I_2, I_3) . mB is a scalar corresponding product of magnetic moment m of the body and strength of the rotating field B . γ denotes the elevation angle the vector $\vec{B}$ makes with its rotation plane (x,y), also called tilt angle. ζ the damping coefficient and t the time. $\theta(t), \phi(t), \psi(t)$ are the rotations of the body defined in Euler rotation matrix in ZXZ scheme.

2. ZXZ Rotation Matrix

$$R = R_z(\phi) R_x(\theta) R_z(\psi), \quad R_z(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad R_x(\beta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \beta & -\sin \beta \\ 0 & \sin \beta & \cos \beta \end{pmatrix}.$$

3. Angular Velocity Vector in Body Frame

$$[\omega]_s = R^T \vec{\omega} \implies \nu = \text{vee}(R^T \vec{R}) = \begin{pmatrix} (R^T \vec{R})_{12} \\ (R^T \vec{R})_{13} \\ (R^T \vec{R})_{21} \end{pmatrix} = \begin{pmatrix} \theta \cos \psi + \phi \sin \psi \sin \theta \\ -\theta \sin \psi + \phi \cos \psi \sin \theta \\ \sin \theta \cos \phi + \psi \end{pmatrix}.$$

4. Rayleigh Dissipation Function

$$\mathcal{D} = \frac{1}{2} \zeta [\dot{\theta}]^2 = \frac{1}{2} \zeta \left(2 \cos(\theta) \dot{\phi}^2 + \dot{\phi}^2 + \dot{\psi}^2 + \dot{\theta}^2 \right).$$

5. Unit Vector of Rotating Magnetic Field

$$\vec{B} = \begin{pmatrix} \sin \omega t \cos \gamma \\ -\cos \omega t \cos \gamma \\ \sin \gamma \end{pmatrix}.$$

6. Unit Vector of Body's Magnetic Moment

$$\hat{m} = B \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} -\cos \phi \sin \theta \\ \sin \phi \sin \theta \\ \cos \theta \end{pmatrix}.$$

7. Moment of Inertia Tensor For This Model

$$I = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}.$$

8. Kinetic and Potential Energy

$$T = \frac{1}{2} \nu^T I \nu = \frac{I_1 (\sin \psi \sin \theta \dot{\phi} + \cos \psi \dot{\phi})^2}{2} + \frac{I_2 (\sin \psi \dot{\theta} - \sin \theta \cos \psi \dot{\phi})^2}{2} + \frac{I_3 (\cos \theta \dot{\phi} + \dot{\psi})^2}{2},$$

$$V = -mB (\hat{m} \cdot \vec{B}) = -mB (\sin \gamma \cos \theta + \sin \theta \cos \psi \cos(\omega t - \phi)).$$

Kinetic Energy of a Body with Isotropic Moment of Inertia

$$T_{\text{iso}} = \frac{I (2 \cos \theta \dot{\phi}^2 + \dot{\phi}^2 + \dot{\psi}^2 + \dot{\theta}^2)}{2}.$$

9. Euler-Lagrange Equations with Damping

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_i} - V \right) - \frac{\partial T}{\partial q_i} = Q_i^{(\text{mech})}, \quad Q_i^{(\text{damp})} = -\frac{\partial \mathcal{D}}{\partial \dot{q}_i}, \quad q_i \in \{\theta, \phi, \psi\}.$$

10. Solve for the Angular Accelerations

$$\begin{aligned} \theta \rightarrow \theta_{\text{dss}}, \quad \dot{\phi} \rightarrow \dot{\phi}_{\text{dss}}, \quad \dot{\psi} \rightarrow \dot{\psi}_{\text{dss}}, \\ \begin{cases} E_{\theta}(\theta, \dot{\phi}, \dot{\psi}, \ddot{\theta}) = 0, \\ E_{\phi}(\ddot{\phi}, \ddot{\psi}) = 0, \\ E_{\psi}(\ddot{\psi}) = 0 \end{cases} \implies \{\theta_{\text{dss}} = f_{\theta}(\ddot{\phi}, \ddot{\psi}), \theta_{\text{dss}} = f_{\psi}(\ddot{\phi}, \ddot{\psi}), \theta_{\text{dss}} = f_{\psi}(\ddot{\phi}, \ddot{\psi})\}. \end{aligned}$$

► B. Basic Definitions and Equations of Motion

1. General rotation matrix in ZXZ convention based on Euler angles θ , ϕ , and ψ

$$R = R_z(\phi) R_x(\theta) R_z(\psi) = \begin{pmatrix} -\sin \phi \sin \psi \cos \theta + \cos \phi \sin \psi & -\sin \phi \cos \psi \cos \theta - \sin \psi \cos \phi \cos \theta & \sin \phi \sin \theta \\ \sin \phi \cos \psi \cos \theta + \cos \phi \sin \psi \cos \theta & -\sin \phi \sin \psi \cos \theta + \cos \phi \cos \psi \cos \theta & -\sin \theta \cos \phi \\ \sin \psi \sin \theta & \sin \theta \cos \psi & \cos \theta \end{pmatrix}.$$

2. Unit vectors $(\vec{B}, \hat{m})$ of the rotating magnetic field and the body's magnetic moment

$$\vec{B} = \begin{pmatrix} \sin \omega t \cos \gamma \\ -\cos \omega t \cos \gamma \\ \sin \gamma \end{pmatrix}, \quad \hat{m} = \begin{pmatrix} \sin \phi \sin \theta \\ \cos \phi \sin \theta \\ \cos \theta \end{pmatrix}.$$

Here ω denotes the angular velocity of $\vec{B}$ around z-axis, and γ the elevation angle of $\vec{B}$ from the xy-plane. The angle ψ does not enter $\hat{m}$ because it corresponds to a self-rotation around the axis of $\hat{m}$.

3. Body's angular velocity vector ν in the lab and in the body's frames

$$\nu_{\text{body}} = \begin{pmatrix} \theta \cos \psi + \phi \sin \psi \sin \theta \\ -\theta \sin \psi + \phi \cos \psi \sin \theta \\ \phi \cos \theta + \psi \end{pmatrix}, \quad \nu_{\text{lab}} = \begin{pmatrix} \theta \cos \phi + \psi \sin \theta \sin \theta \\ \theta \sin \phi - \psi \cos \theta \sin \theta \\ \psi \cos \theta + \phi \end{pmatrix}.$$

4. Equations of motion based on Euler angles θ , ϕ and ψ , where I_1, I_2 are radial and axial components of the moment-of-inertia tensor of an axisymmetric body, and ζ is the damping coefficient

$$I_1 \ddot{\theta} = mB (-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

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$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right) I_1^2 I_2 \left((-\sin^2 \theta \cos \psi \dot{\phi}^2 + \cos \psi \dot{\theta}^2) \sin^4 \psi + \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 3 \sin^4 \psi \sin \theta \dot{\phi} \dot{\theta} - 2 \sin^4 \psi \sin^2 \theta \cos \psi \dot{\phi}^2 - 2 \sin^4 \psi \sin \psi \cos \psi \dot{\theta}^2 + \sin^2 \psi \sin \theta \dot{\phi} \dot{\psi} + \sin^2 \psi \sin^2 \theta \right.$$

$$\left. I_1 \dot{\psi} \right) = mB \left((-I_1 \sin \gamma \sin \theta + (-I_1 + I_2) (\sin \gamma \sin^2 \psi \sin \theta + \sin(\omega t - \phi) \sin \psi \cos \gamma \cos \psi) + (I_1 \sin^2 \psi + I_2 \cos^2 \psi) \cos \gamma \cos(\omega t - \phi) \cos \theta) + \zeta \left((-I_2 \dot{\phi} + (-I_1 + I_2) \dot{\psi} \sin \theta \cos \phi + \right.$$

$$\left. -I_1 \dot{\theta} \right) mB (-I_1 - I_2) \sin \gamma \sin \psi \sin \theta \cos \psi + (-I_1 - I_2) \sin \psi \cos(\omega t - \phi) \cos \psi \cos \theta + (I_1 - (I_1 - I_2) \sin^2 \psi) \sin(\omega t - \phi) \cos \theta) + \zeta \left((I_1 - I_2) \sin \psi \cos \psi \dot{\theta} + \right.$$

$$\left. I_1 \dot{\theta} \right$$